

RIYA MITRA

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Portfolio Website: <https://riya-mitra.github.io/portfolio/>

EDUCATION

UNIVERSITY OF PENNSYLVANIA, Bachelor of Science and Engineering

May 2025

- **Bachelor of Science & Engineering** in Computer Science and Minor in Mechanical Engineering and Engineering Entrepreneurship, **GPA: 3.4**
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Computer Vision & Computational Photography, Scalable Systems & Cloud Computing, Human Computer Interaction, Engineering Entrepreneurship, Negotiations, Economics
- **Organizations:** Society of Women Engineers, Penn Innovators in Business, Wharton Undergraduate Entrepreneurship, Theta Tau Pre-Professional Engineering Fraternity

WORK EXPERIENCE

NVIDIA, DRIVE OS Technical Program Manager Intern

May 2024 – August 2024

- Conducted market research and analysis to identify inefficiencies in program management workflows, developing automation products that optimized daily tasks by 2-3 hours, directly contributing to key feature releases (DRIVE OS 6.0.9.2 and 6.0.9.3) and aligning with critical KPIs.
- Designed and launched a LLM chatbot powered by RAG that ingests and indexes data from Jira, Jama, and other databases using Langchain to provide program managers with rapid, efficient access to critical information, enhancing productivity across the team.
- Developed a Python-based platform used by cross functional teams that automates the analysis of over 500,000 work product files from Perforce, reducing manual processing time by 90% and adding enhanced data accuracy.
- Collaborated with the Documentation PM to re-engineer and automate the release documentation workflow by developing a Python platform that integrates multiple APIs and NVIDIA's internal tools, reducing the process duration from 2-3 days to minutes, improving efficiency and accelerating customer delivery timelines.

MODLAB ROBOTICS, Robotics Software Research Intern

May 2023 – May 2024

- Designed and prototyped an innovative snap-back linking mechanism for an advanced tunnel-traversing robot, achieving a 45% increase in traversal speed and streamlining operational efficiency by 30%.
- Collaborated with cross-functional teams to subject the snap back linking mechanism to rigorous stress testing, certifying its durability withstanding over 1,000 cycles without any mechanical failures.
- Developed and implemented a Python simulation for a dual rotating robot that models its motion and calculates torque, demonstrating 90% accuracy in emulating real-world scenarios.
- Developed a Python algorithm for the linking mechanism to dynamically process varying size inputs, optimizing software efficiency and accurately predicting maximum achievable tunnel dimensions.

ORIDEN POWER, Engineering Intern

June 2022 – August 2022

- Led data-driven feasibility studies for 30+ projects to optimize solar design configurations, maximizing energy yield and identified 25 new potential project sites.
- Presented an in-depth research project on new solar technology which increases energy yield by 24% and applied new software models to determine energy production and viability of new technology.
- Engineered a Java application to output the most cost-effective solar panel configurations which increased efficiency and reduced material cost by 30%.

SKILLS & PROJECTS

Java, Python, HTML, CSS, JavaScript, React, SQL, RAG, OpenCV, MATLAB, Arduino, Jira, Agile, Figma

Insta-lite: Javascript, RAG, LLMs, Kafka, SQL, React, Java Spark, AWS RDS, ChromaDB,

- “Instagram” social media web platform for users to share photos, post status updates, chat with other users, chat with an LLM, and get real time news from X using Kafka.

Bloom: React, Javascript, Figma, User market research, Prototyping

- Gamified, multiuser roommate organizer app to manage and assign household chores while maintaining a virtual a flower garden